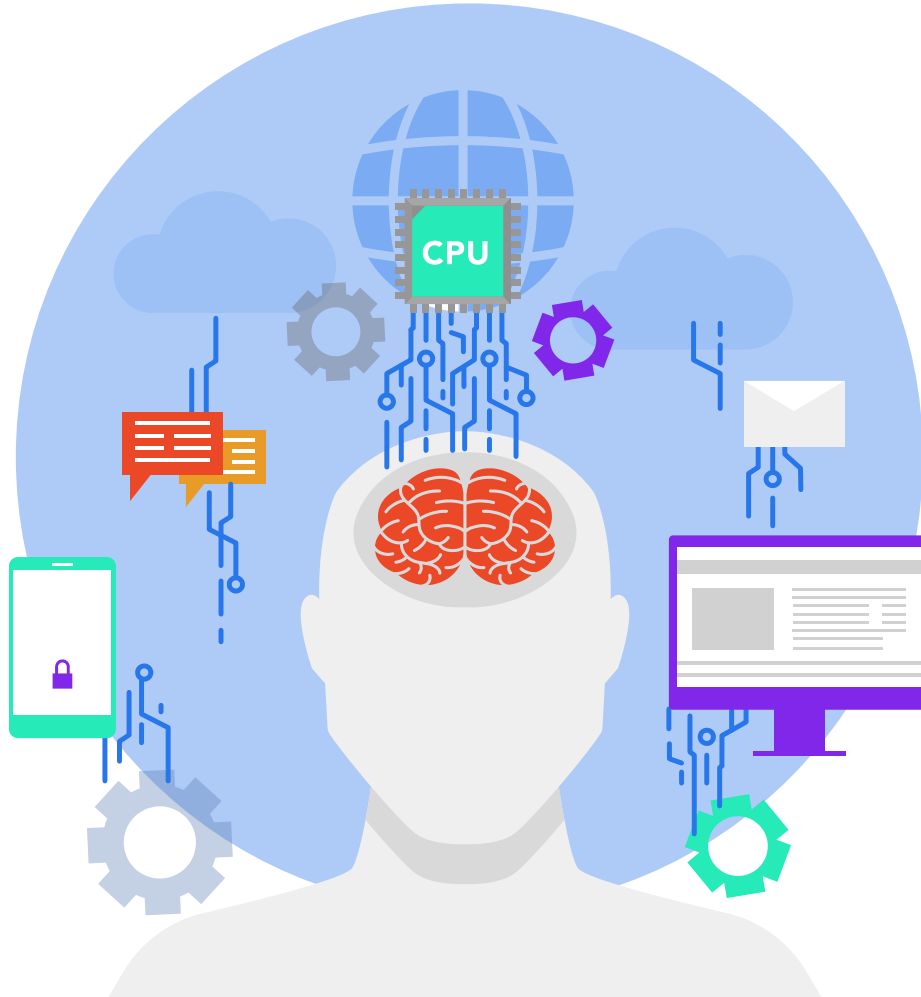
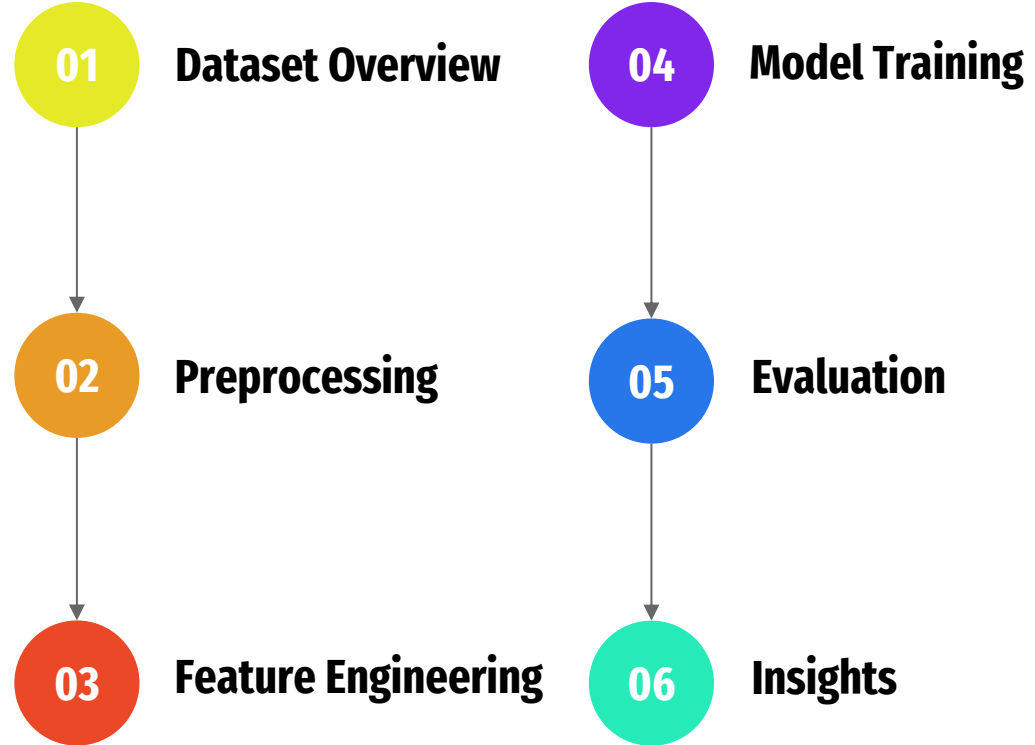
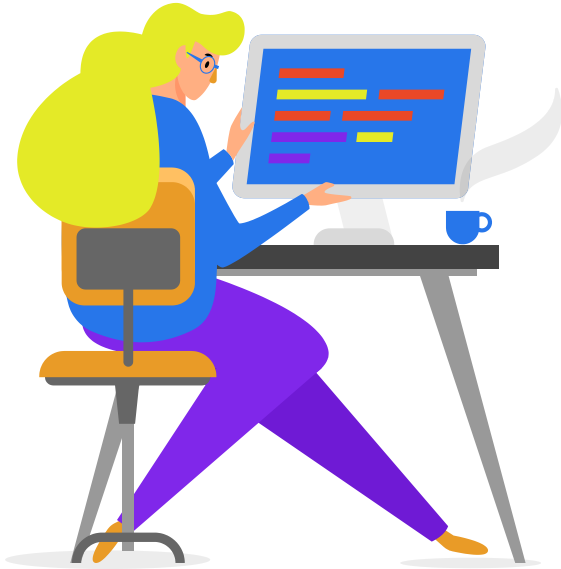




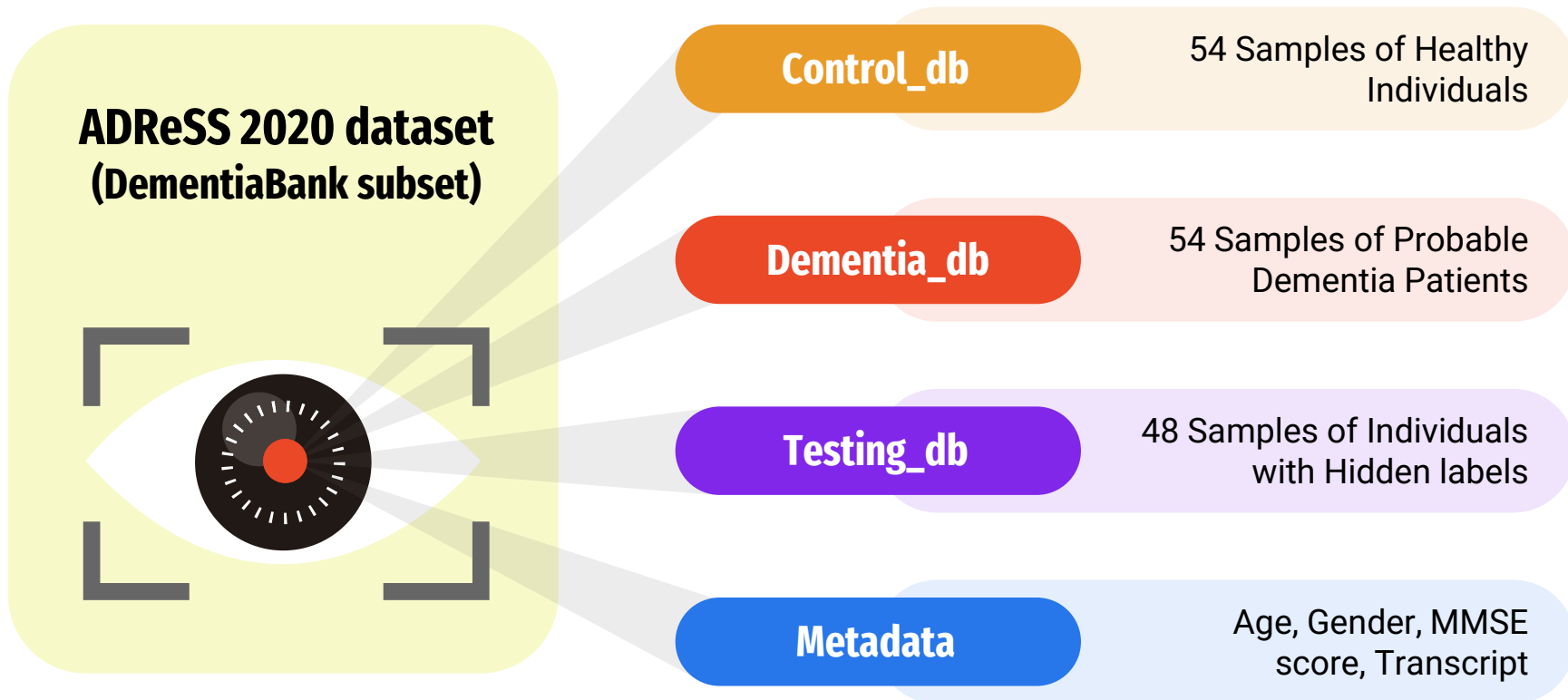
Detecting Dementia Using Natural Language Processing

B M Nafis Fuad

Outline

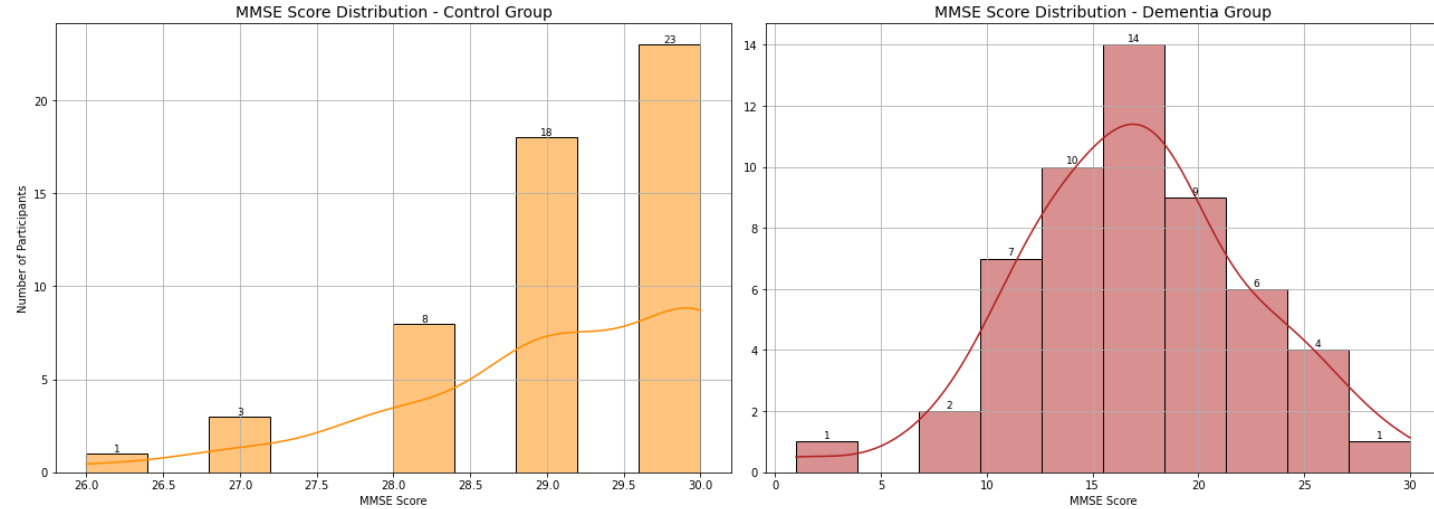


Dataset Overview



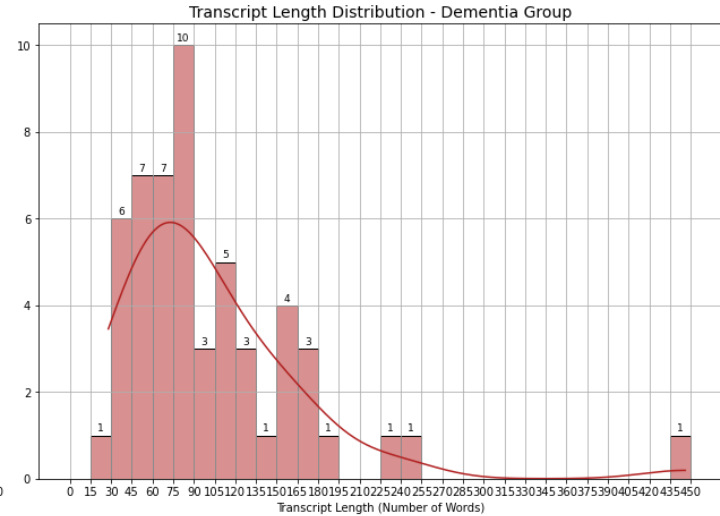
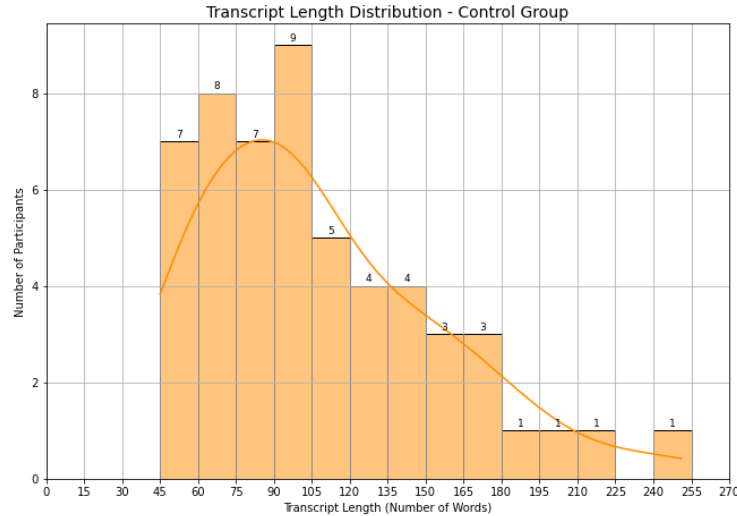
Dataset Overview

Comparison of MMSE Score Distribution Between Control and Dementia Groups

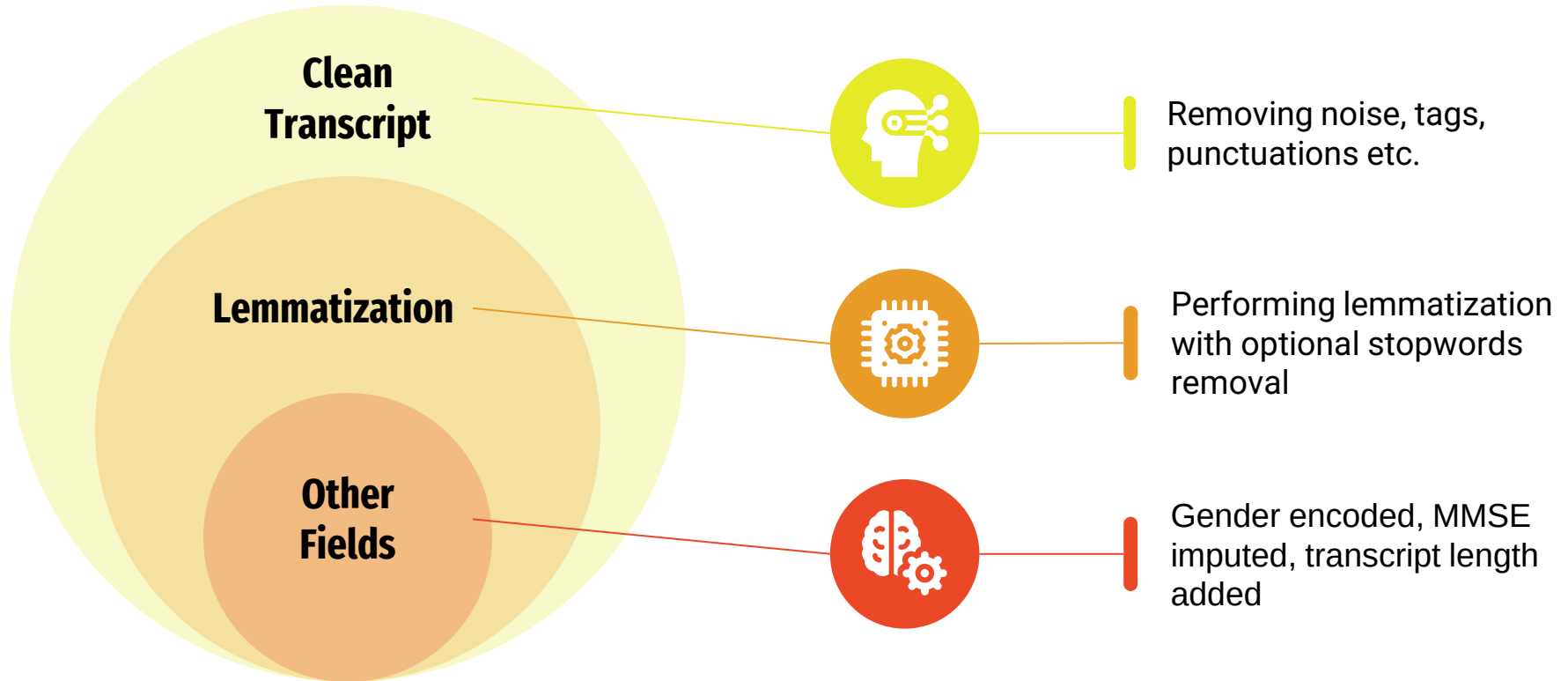


Dataset Overview

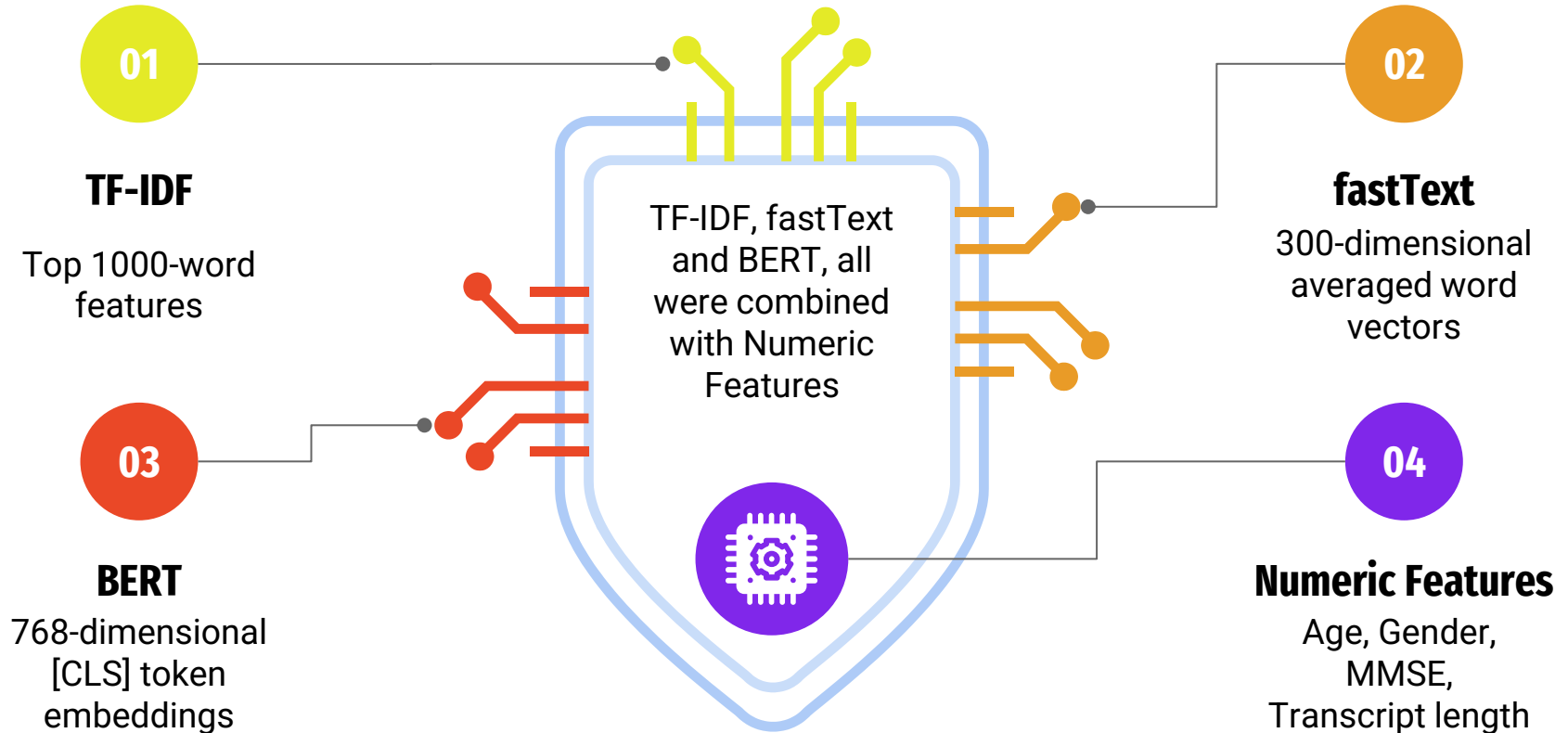
Comparison of Transcript Length Distribution Between Control and Dementia Groups



Preprocessing



Feature Engineering



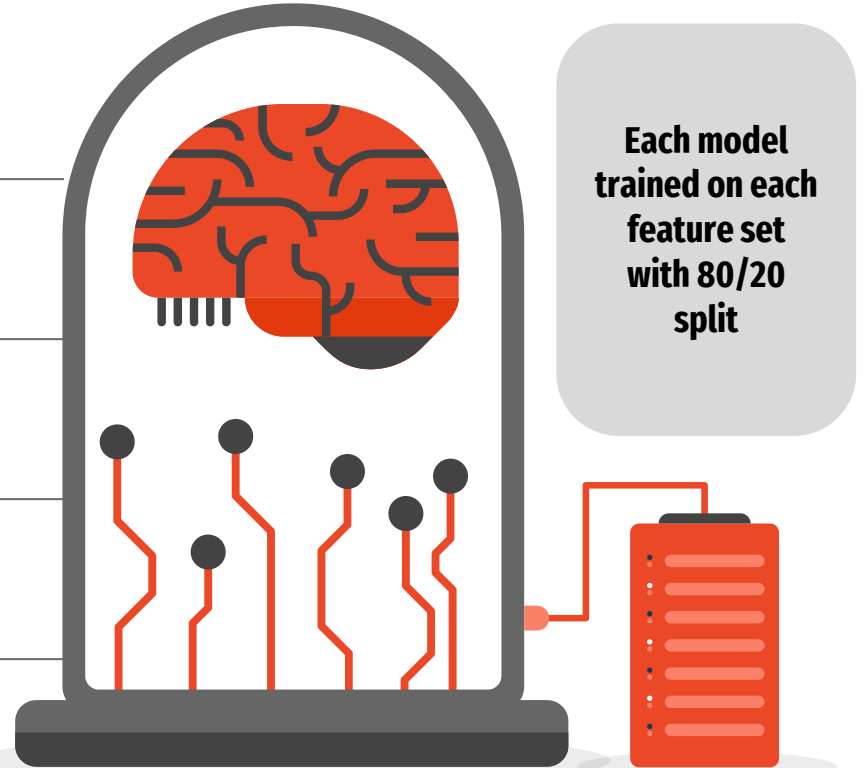
Model Training

01 Logistic Regression

02 SVM

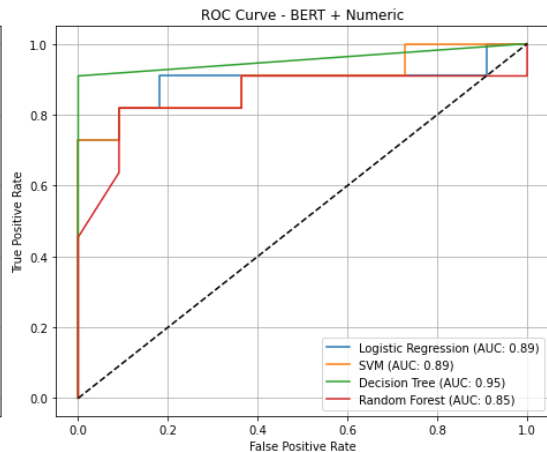
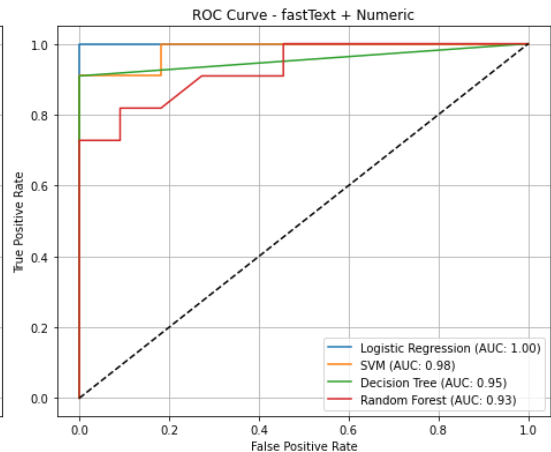
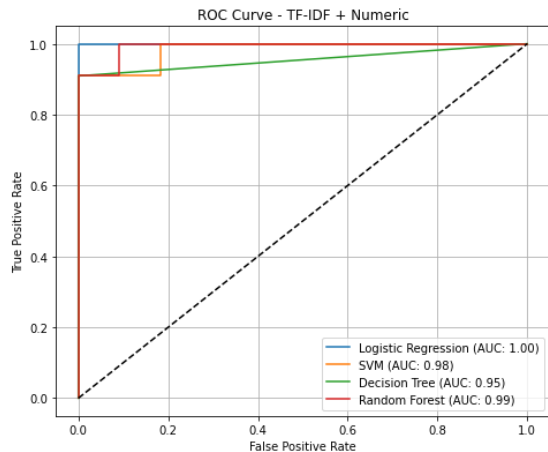
03 Decision Tree

04 Random Forest

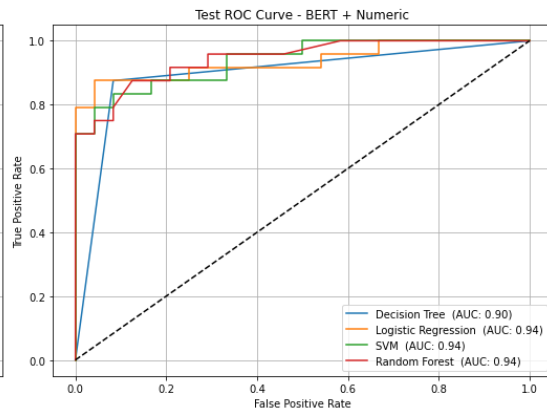
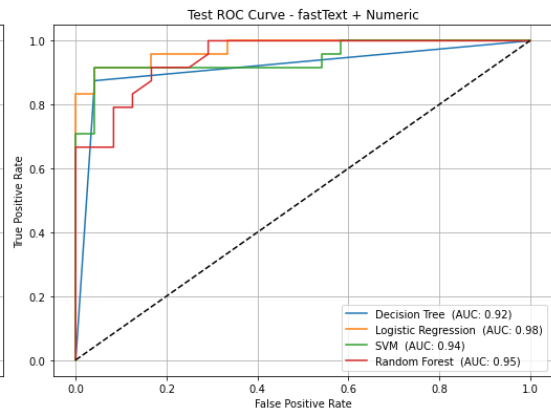
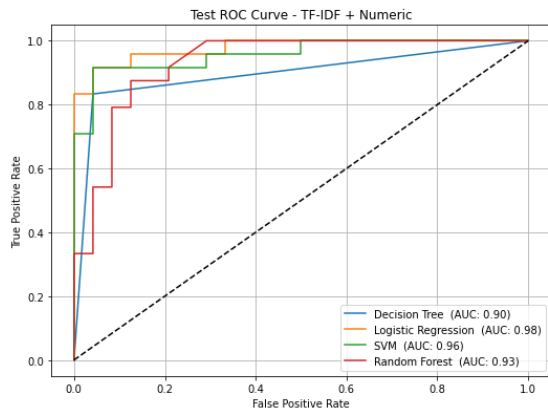


Evaluation

ROC Curves for All Models by Feature Set



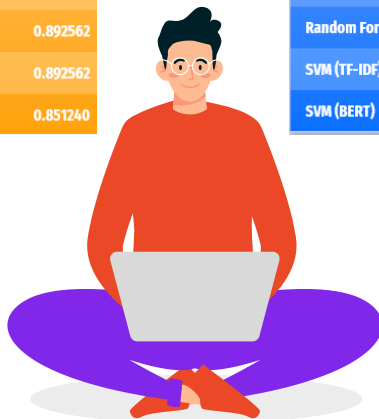
ROC Curves for All Models (Test Set)



Evaluation

Model+Feature	Accuracy	Precision	Recall	F1	ROC AUC
Decision Tree (TF-IDF + Numeric)	0.954545	1.000000	0.909091	0.952381	0.954545
Decision Tree (fastText + Numeric)	0.954545	1.000000	0.909091	0.952381	0.954545
Decision Tree (BERT + Numeric)	0.954545	1.000000	0.909091	0.952381	0.954545
Logistic Regression (TF-IDF + Numeric)	0.909091	1.000000	0.818182	0.900000	1.000000
SVM (TF-IDF + Numeric)	0.909091	1.000000	0.818182	0.900000	0.983471
Logistic Regression (fastText + Numeric)	0.909091	1.000000	0.818182	0.900000	1.000000
SVM (fastText + Numeric)	0.909091	1.000000	0.818182	0.900000	0.983471
Random Forest (TF-IDF + Numeric)	0.863636	1.000000	0.727273	0.842105	0.991736
Random Forest (fastText + Numeric)	0.818182	0.769231	0.909091	0.833333	0.929752
Logistic Regression (BERT + Numeric)	0.818182	0.818182	0.818182	0.818182	0.892562
SVM (BERT + Numeric)	0.818182	0.818182	0.818182	0.818182	0.892562
Random Forest (BERT + Numeric)	0.727273	0.692308	0.818182	0.750000	0.851240

Model+Feature	Accuracy	Precision	Recall	F1	ROC AUC
Decision Tree (fastText)	0.916667	0.954545	0.875000	0.913043	0.916667
Logistic Regression (BERT)	0.916667	0.954545	0.875000	0.913043	0.935764
Decision Tree (BERT)	0.895833	0.913043	0.875000	0.893617	0.895833
Decision Tree (TF-IDF)	0.895833	0.952381	0.833333	0.888889	0.895833
Logistic Regression (TF-IDF)	0.895833	1.000000	0.791667	0.883721	0.977431
Logistic Regression (fastText)	0.895833	1.000000	0.791667	0.883721	0.975694
Random Forest (TF-IDF)	0.854167	0.904762	0.791667	0.844444	0.930556
SVM (fastText)	0.854167	1.000000	0.708333	0.829268	0.944444
Random Forest (fastText)	0.833333	0.863636	0.791667	0.826087	0.947917
Random Forest (BERT)	0.833333	0.900000	0.750000	0.818182	0.943576
SVM (TF-IDF)	0.833333	0.944444	0.708333	0.809524	0.958333
SVM (BERT)	0.833333	0.944444	0.708333	0.809524	0.937500



Insights

01

Decision Trees performed consistently well across all feature sets, particularly in fastText and BERT configurations

Logistic Regression with BERT generalizes better on test data despite slightly lower training performance

02

03

TF-IDF features paired with classical models (LR, SVM) still offer competitive results

Thank You